Xiaotong (Kary) Fang

KaryFang9 | PirayLab | XiaotongFang | Kary's Website | fangxiao@usc.edu | 2137051774

Summary

PhD Candidate in Computational Neuroscience at USC with experience developing machine learning, forecasting, and causal inference systems across large-scale business and behavioral datasets. Built predictive models on datasets ranging from 500K to 10M+ records and developed probabilistic modeling frameworks for uncertainty estimation, sequential decision-making, and impact measurement. Experienced in translating complex modeling outputs into actionable business insights through scalable data pipelines, experimentation, and stakeholder-facing analytics.

Focus Areas: time-series modeling, sequence models, Bayesian inference, uncertainty estimation, causal inference

Skills

Programming	Python, SQL, R, MATLAB, Java, JavaScript, Git
ML	XGBoost, LightGBM, Random Forests, Neural Networks, Sequence Modeling, Bayesian Modeling, Reinforcement Learning
Statistics & Experimentation	Causal Inference, A/B Testing, Hypothesis Testing, Regression, Factor Analysis, Statistical Inference, Uncertainty Quantification
Data & MLOps	Scikit-Learn, PyTorch, REST APIs, Feature Engineering, Model Deployment, Data Pipelines
Community	Teaching Assistant (USC); Treasurer, Young Researchers Program

Publications

Fang, Xiaotong and Payam Piray (2025). "Inferring the causes of noise from binary outcomes: A normative theory of learning under uncertainty". In: *Psychological Review*. URL: https://psycnet.apa.org/fulltext/2028-04912-001.html.

Zhang, X., S. Ivanovic, B. Moore, X. Fang, and D. Song (2025). "Topological Masking and Attention for Universal Decoding of Spiking Neural Dynamics". In: *NeurIPS '25*. URL: https://openreview.net/forum?id=A6WQTfLqJ7.

Fang, Xiaotong and Payam Piray (2026). "Dissociating volatility and stochasticity reveals transdiagnostic computational signatures of psychopathology". In: *bioRxiv*. DOI: 10.64898/2026.05.22.727329. URL: https://www.biorxiv.org/content/early/2026/07/05/2026.05.22.727329.

Work & Research Projects

Google (BDS) — R&D Scalable Uncertainty Quantification for RMSN May 2026 – Aug 2026

- Developed scalable uncertainty quantification methods for RMSN, a deep causal inference model measuring sales impact across 20M+ advertiser time-series records over 9 quarters.
- Built a bootstrap confidence interval framework by randomizing dataset splits and executing the causal inference pipeline end-to-end for empirical uncertainty benchmarking.
- Identified that bootstrap-based intervals primarily captured epistemic uncertainty in the existing mean-only model, motivating model-level variance estimation.
- Replaced the LSTM decoder with Bayesian neural network layers to generate both mean and variance outputs for uncertainty-aware counterfactual prediction.
- Implemented custom Bayesian training steps, probabilistic loss functions, Monte Carlo inference, and transfer-learning strategies to stabilize Bayesian decoder training.
- Integrated code changes into the active RMSN pipeline with validation tests to support smooth cross-team handoff, reproducible evaluation, and future experimentation.
- Partnered with data engineering, data science, and business teams to translate confidence interval estimates into decision support for sales performance evaluation and resource allocation.

Beijing Topjoy Technology Co. Ltd.(DS) — Large-Scale Churn Prediction May 2021 – Sep 2021

- Built churn prediction models on 10M+ user behavior logs using SQL, Python, Random Forest, and XGBoost
- Engineered time-series and behavioral decay features to capture user activity cycles and trend shifts
- Improved model performance to AUC 0.84 (+18% vs baseline) through feature selection and hyperparameter tuning
- Deployed models via REST API to support batch inference and retention strategy analysis
- Conducted feature importance analysis to enhance model interpretability and inform operational decisions

Piray Lab(RA) — Sequential Decision Modeling Oct 2025 – Present

- Built RNN-based sequence modeling framework with Knowledge Distillation for joint prediction of choice and reaction time
- Evaluated models on 4,000+ users to characterize dynamic decision processes
- Designed composite loss functions balancing classification accuracy and temporal modeling performance
- Analyzed latent state trajectories to improve interpretability and stability

Piray Lab(RA) — Uncertainty Decomposition and Latent State Modeling Aug 2023 – Oct 2025

- Designed and deployed a web-based behavioral experiment platform (JavaScript + Python), collecting sequential decision data from 4,600+ participants
- Built hierarchical Bayesian and state-space models to disentangle stochasticity and environmental volatility
- Performed parameter estimation using Variational Inference and Maximum Likelihood Estimation (MLE) for stable latent variable inference
- Applied factor analysis to extract stable computational features capturing individual behavioral differences
- Developed interpretable modeling frameworks to analyze the impact of different uncertainty sources on decision behavior

Neural Decoding and Interface Lab(RA) — Cross-Subject Neural Decoding Aug–Dec 2022

- Developed PyTorch decoder with topological masking and attention modules.
- Achieved +15% accuracy across 59 subjects via transfer learning and domain adaptation.
- Implemented reproducible training and evaluation pipelines.

Booking Forecasting and Prediction Modeling May 2025

- Built an end-to-end pipeline on 520k booking records (EDA → feature engineering → imputation → scaling → training)
- Designed time-window and hierarchical aggregation features to capture historical behavior patterns
- Benchmarked XGBoost, LightGBM, and linear models under multiple cross-validation strategies
- Improved MAE/RMSE by 15% via feature enhancement and loss optimization
- Applied SHAP for feature attribution and implemented quantile regression for interval forecasting and risk estimation

Education

University of Southern California, Dornsife GPA: 3.71/4.0
PhD in Computational Neuroscience | Enrolled 2022, Expected May 2027
Coursework: CSCI: Deep Learning and its Applications (Pytorch, LLM), DSCI: Machine Learning (Data Mining), EE: Computing Principles (C++, Bazel), ECE: Probability, EE: Data Analysis & Neurotech Design (Pytorch), PSYC: Decision Neuroscience (Bayesian Modeling)

Emory University, Emory College GPA: 3.75/4.0
BS in Quantitative Statistical Science and Computer Science | 2018 - December 2021